HerSpark Ideathon 2025

Thakur College of Engineering and Technology (TCET), Mumbai

Title – SafeHerNet

Track 2 -Cyber Safety & Digital Empowerment

ABSTRACT

Cyber Safety & Digital Empowerment

In today's digital world, women are increasingly active online—learning, working, socializing, and leading. But with this empowerment comes vulnerability. Identity theft, cyberbullying, deepfake misuse, and online harassment affect women more harshly than men, often silencing their voices and limiting their digital confidence.

Most existing solutions are either too technical, reactive, or not designed with women in mind. What's missing is a holistic, women-centric platform that doesn't just protect—but also empowers.

That's why we created SafeHerNet—an AI-powered Cyber Safety & Empowerment Platform designed to give women both protection and confidence in the digital space.

How SafeHerNet Helps Women ?:

- Real-time AI Shield: Stops phishing, malicious messages, and even deepfake/voice-cloning scams before they harm.
- Digital Literacy Hub: Fun, gamified lessons in multiple languages, using AR and interactive quizzes to teach safe digital habits.
- Secure Support System: Anonymous reporting and SOS channels that instantly connect victims to trusted NGOs, cyber cells, and mental health experts.
- Digital Footprint Visualizer: A simple dashboard showing how much of your personal data is exposed—and how to fix it.
- Emotion-Aware Chatbot: A friendly AI companion that can sense stress in your messages, listen, and guide you toward safety or professional help

- Women Cyber Leader Network: A safe, supportive community where women mentor each other, share experiences, and build digital leadership.

What Makes It Different:

SafeHerNet is not just a cybersecurity tool—it's a movement. By merging cutting-edge AI (deepfake detection, NLP, blockchain) with human-centered design (mentorship, emotional support, gamified learning), it transforms fear into confidence and victims into leaders.

CASE STUDIES:

- A Delhi student victimized by a deepfake scam in 2023 shows why media authenticity checks are urgent.
- 8 in 10 women hesitate to report online harassment due to stigma (Norton, 2022)—our anonymous reporting breaks this silence.
- The Cambridge Analytica data breach revealed the dangers of data misuse—our footprint visualizer helps women regain control of their digital identities.

The Impact We Envision:

SafeHerNet will help women feel safer, smarter, and stronger online. More than just fighting cyber threats, it builds a generation of women cyber leaders—resilient, confident, and capable of protecting themselves and others.

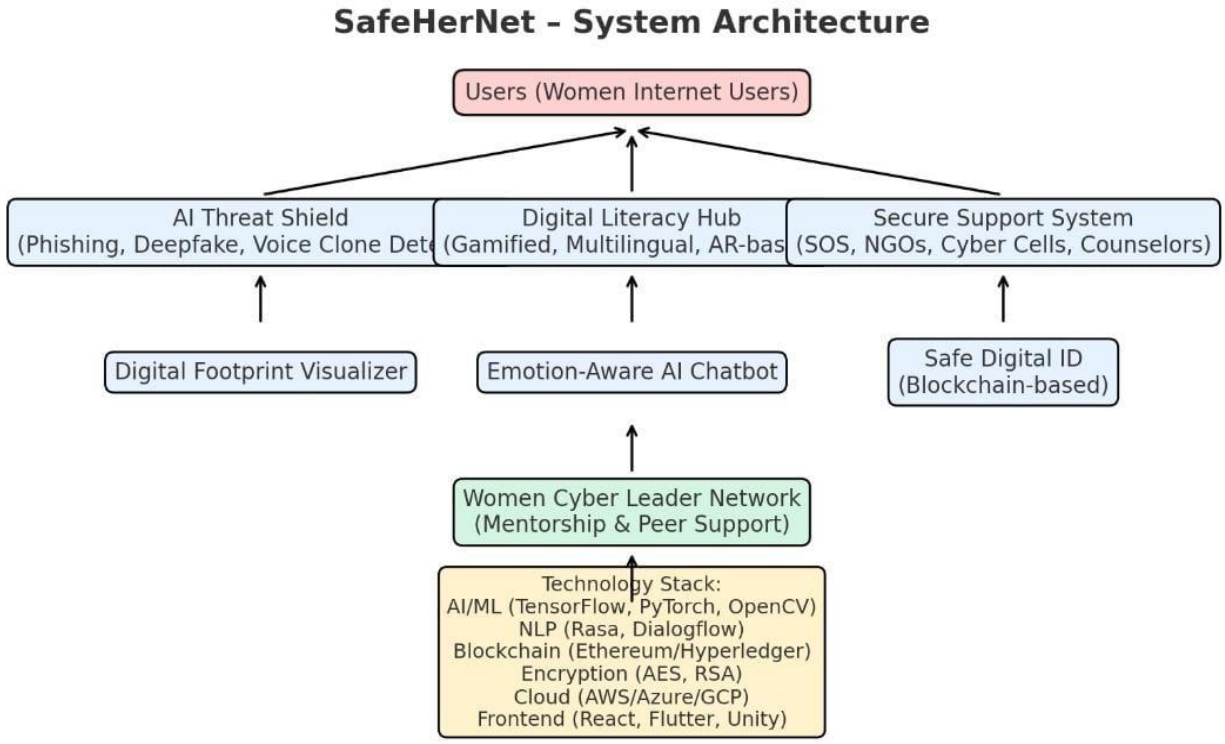
Our vision is bold yet simple: to turn today's vulnerable users into tomorrow's empowered changemakers, and to make cyberspace safe, inclusive, and empowering—for all.

Feasibility & Technology Stack

SafeHerNet is technically achievable today using existing tools:

- AI & Machine Learning: TensorFlow / PyTorch for phishing detection, OpenCV + deep learning for deepfake checks.
- NLP & Chatbot: Rasa / Dialogflow to analyze abusive messages and detect emotional stress.
- Cybrsecurity: Blockchain (Hyperledger/Ethereum private chains) for Safe Digital ID; AES/RSA encryption for secure reporting.

- Frontend & AR Learning: Flutter/React for app development; Unity/ARKit for gamified literacy modules.
- APIs & Cloud: "HaveIBeenPwned" for breach checks; AWS/Azure for scalable, secure infrastructure.
- Analytics: Kibana/Power BI dashboards to track threats, user safety scores, and literacy progress.



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